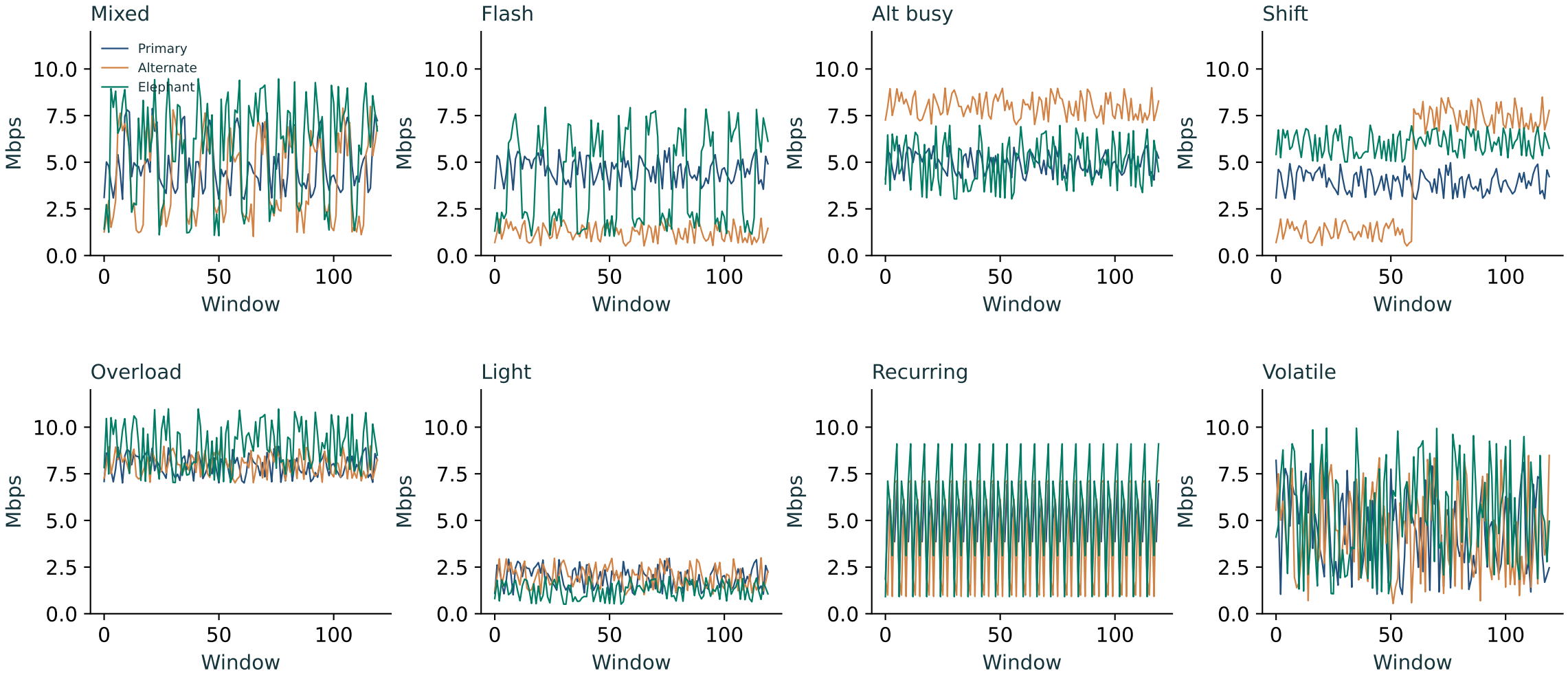
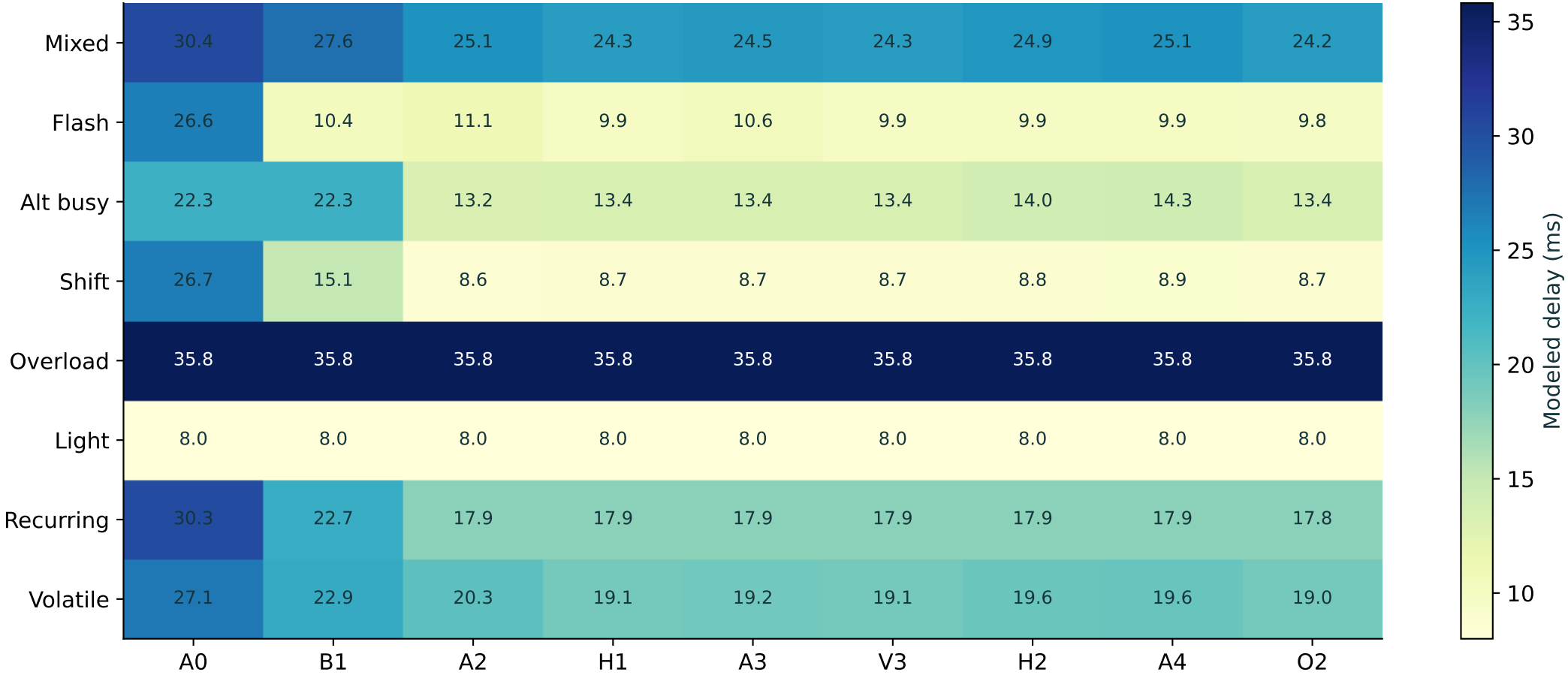
Generated traffic on eight held-out scenarios



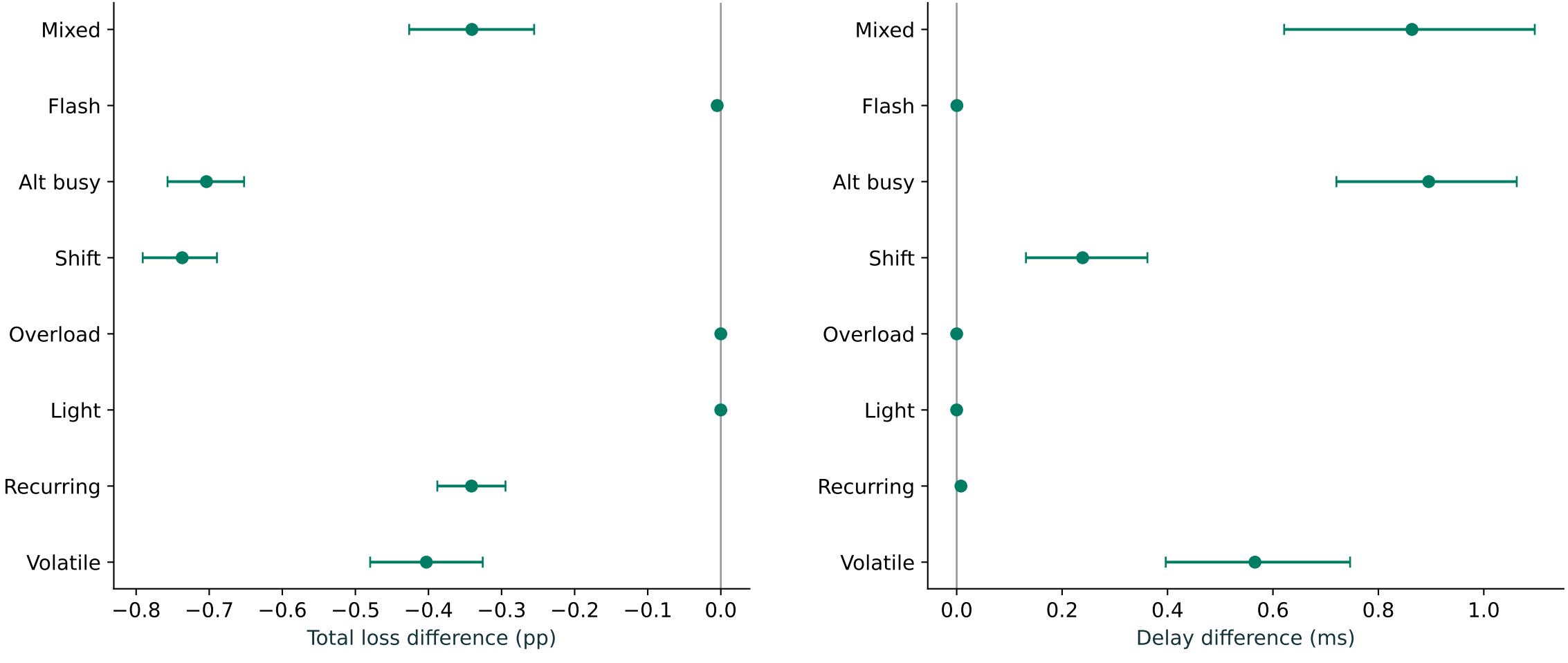
One seed shown (1900). 10 paired seeds × 120 windows × 8 scenarios; capacities 10 Mbps per path. Demand traces are synthetic and hash-verified.

Modeled delay by scenario and controller



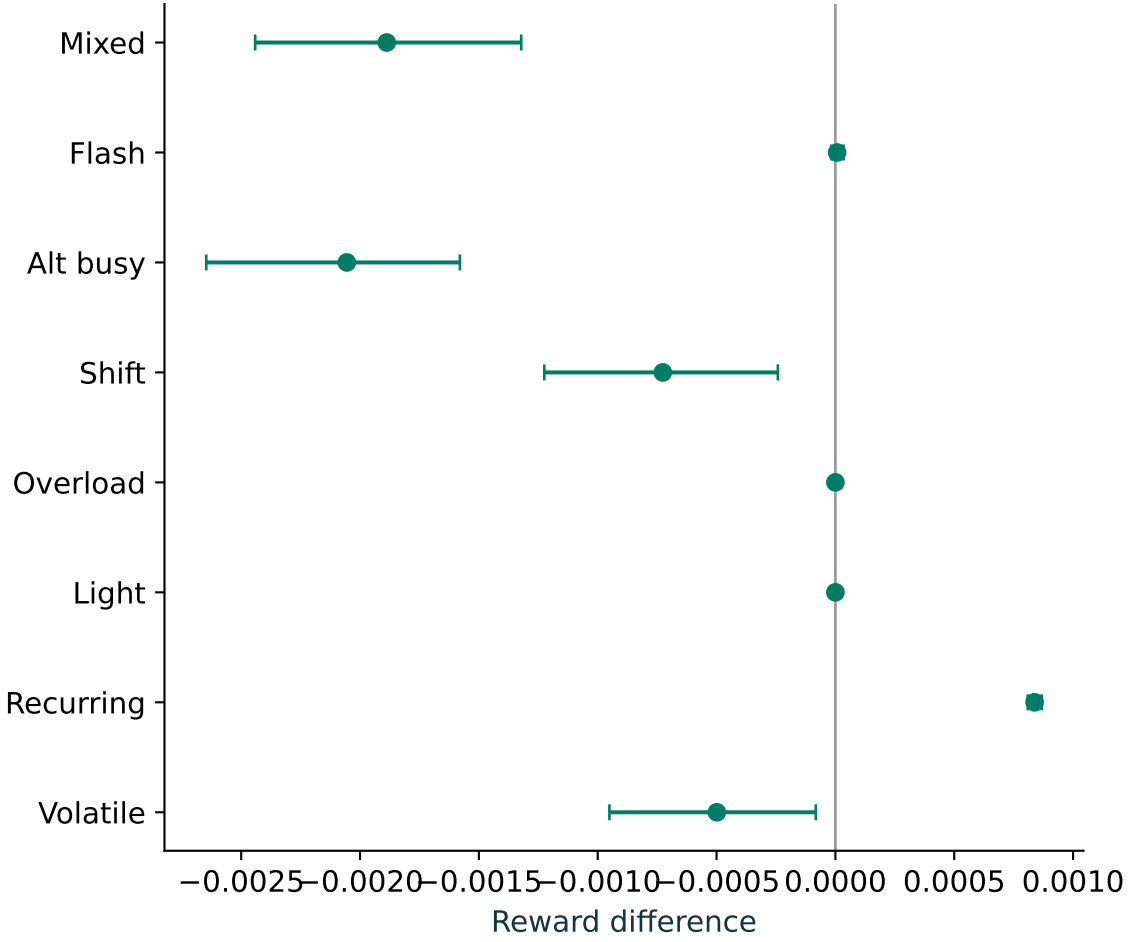
86,400 controller windows, 9,600 unique workload windows. Queue-based fluid delay; no packet RTT or physical-network measurements.

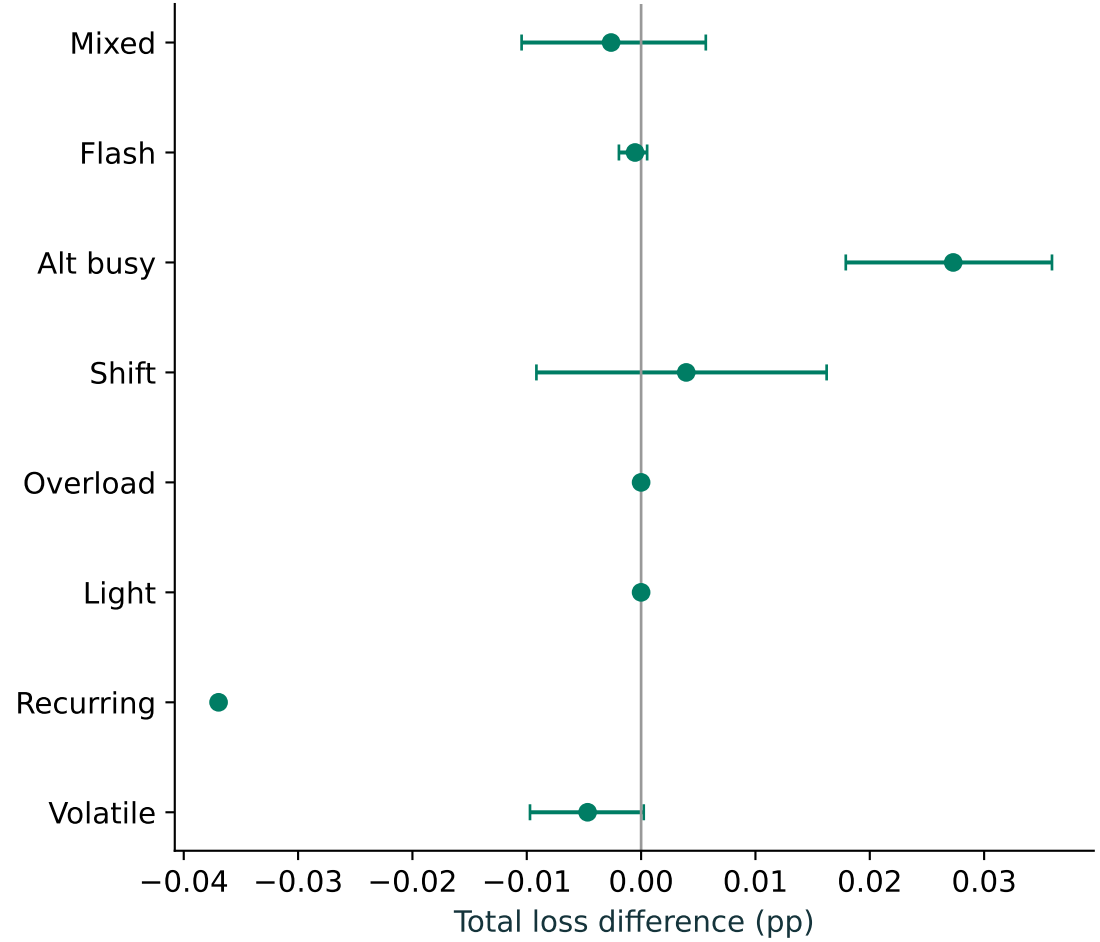
A4 versus H1: loss improves, delay has a tradeoff



A4 minus H1 with paired 95% seed bootstrap intervals (4,000 draws). Negative favors A4. Improvements over H1 include capacity-aware meter logic, not only memory.

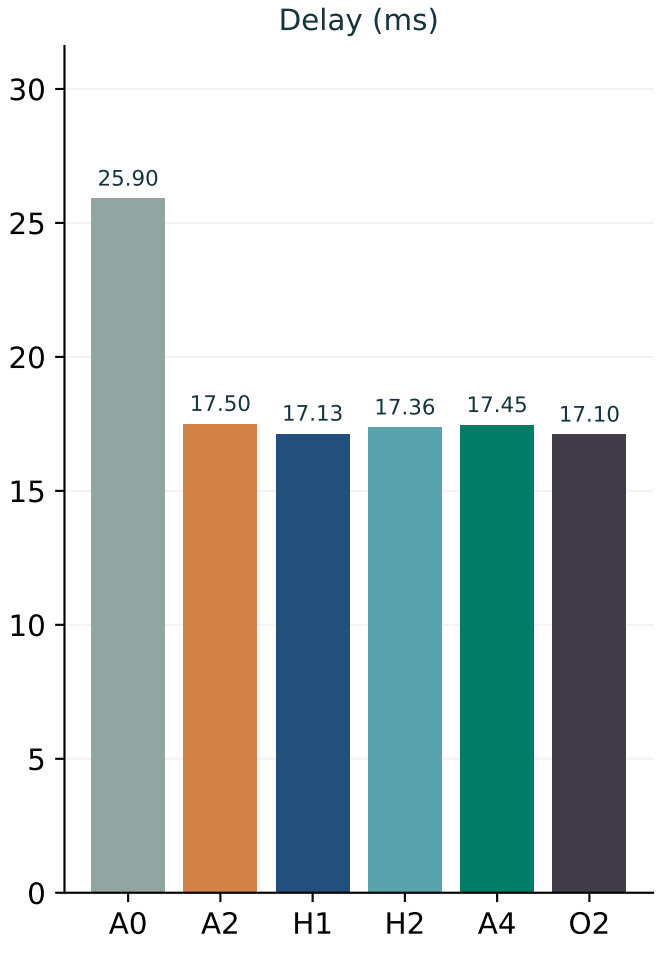
Matched H2 ablation isolates episodic parameter selection

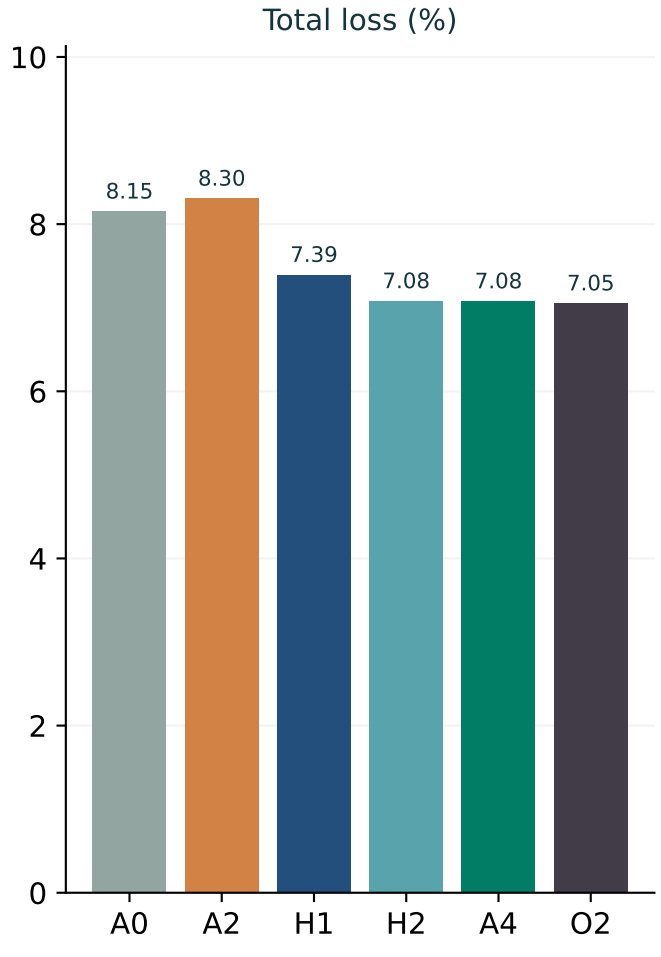


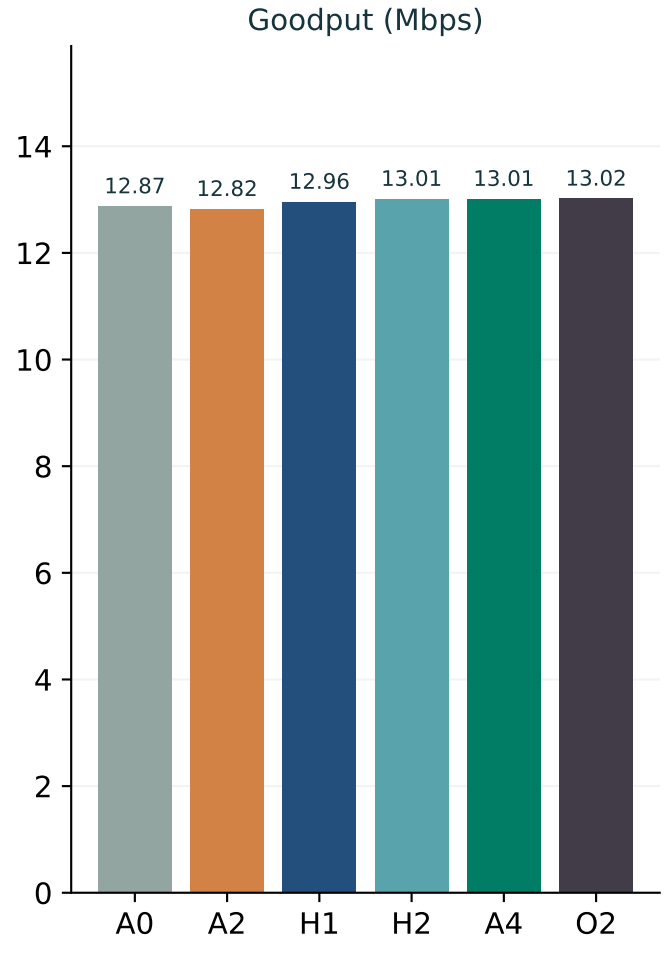


H2 shares the capacity-aware meter and uses a fixed 0.1 Mbps headroom. A4 explores 0, 0.1 and 0.3 Mbps, then selects using observed prior reward. Cold-start exploration stays in results.

Delay, loss and delivered goodput together

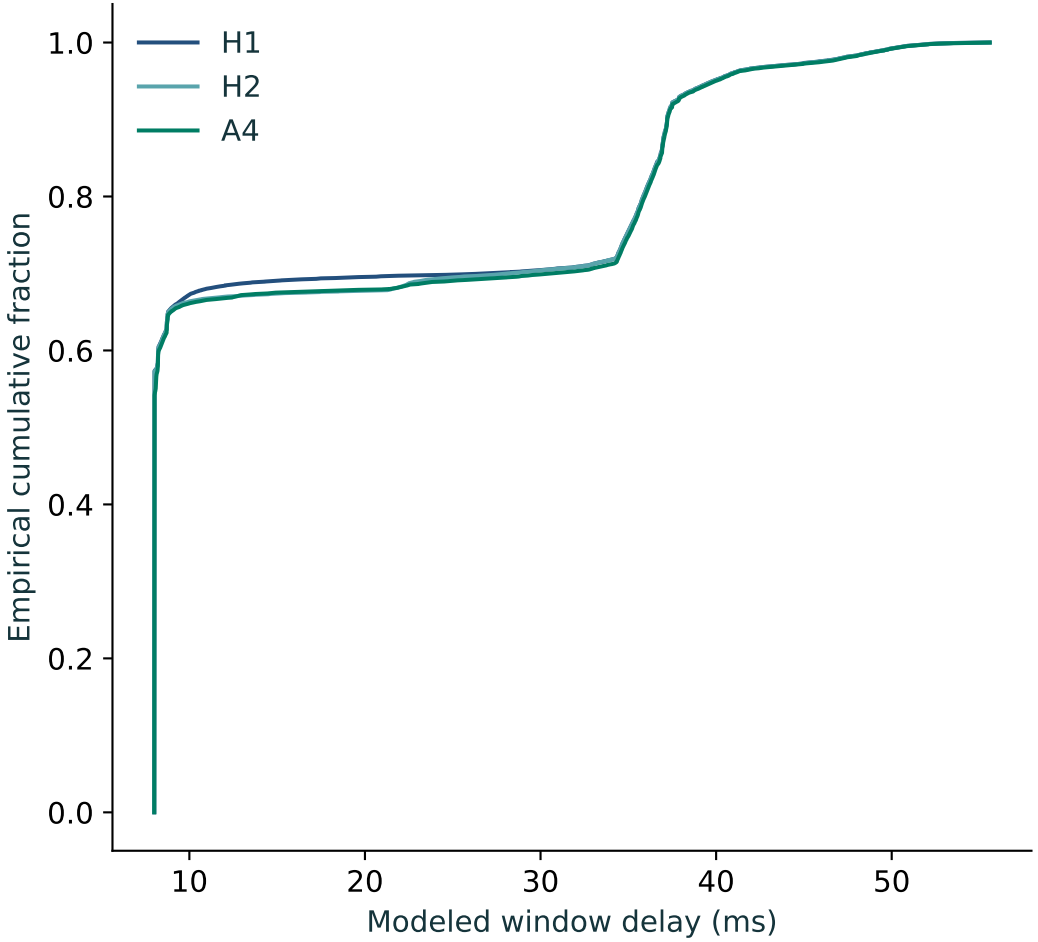


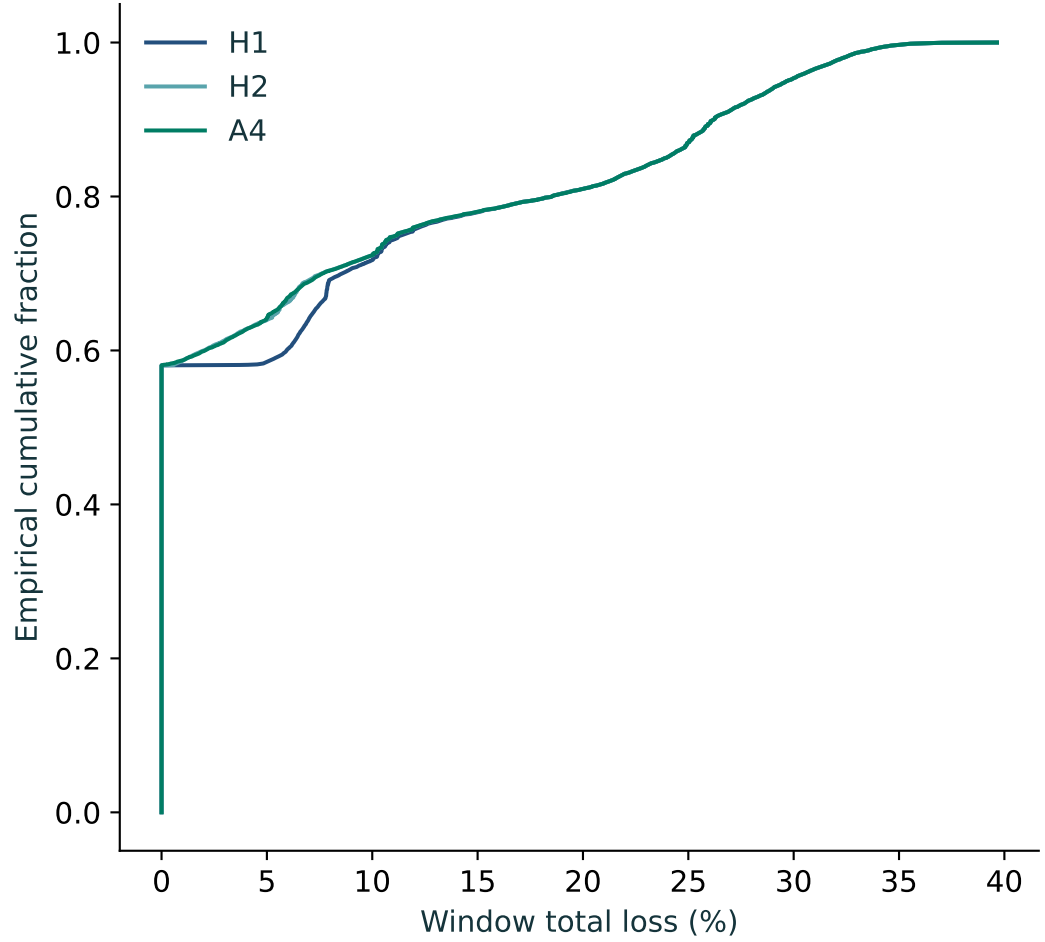




Equal weight per scenario and seed. All loss includes deliberate meter drops and buffer overflow. H1 and previous v3 A4 match; H2 and A4 differ in parameter selection.

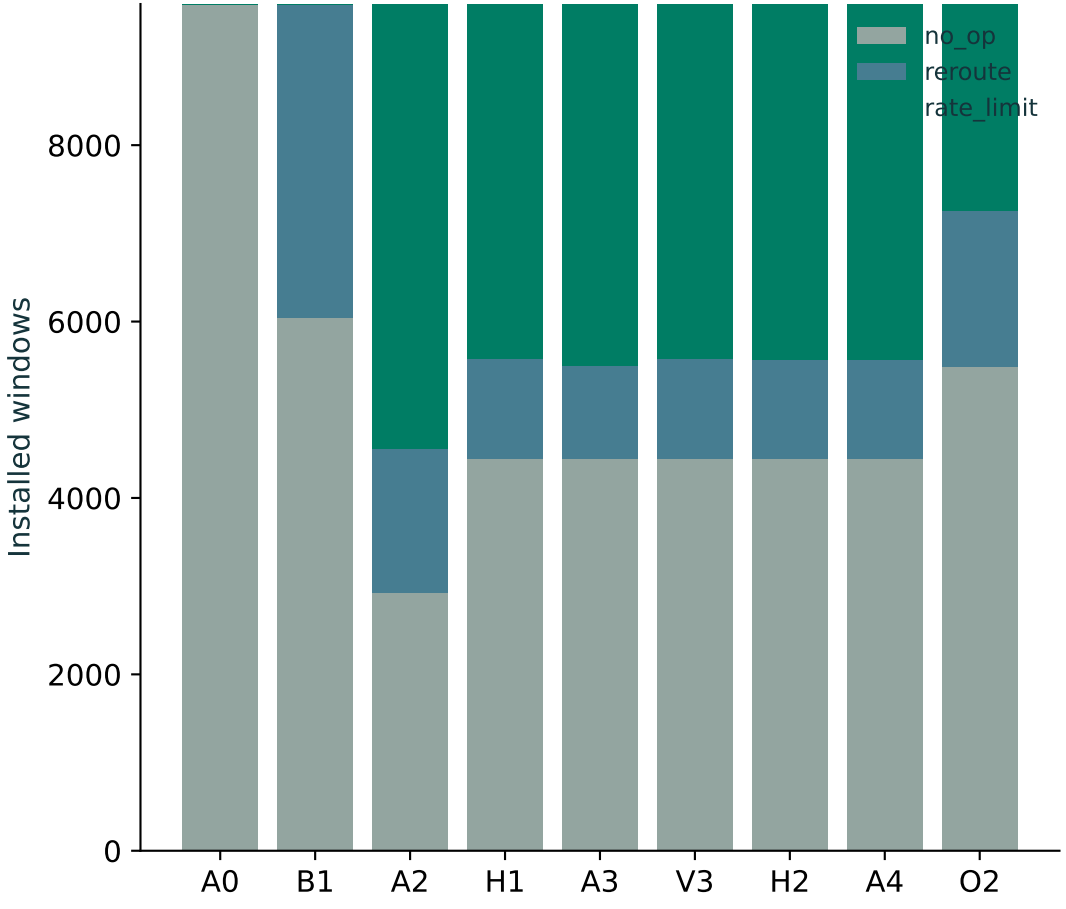
Distribution of modeled window outcomes

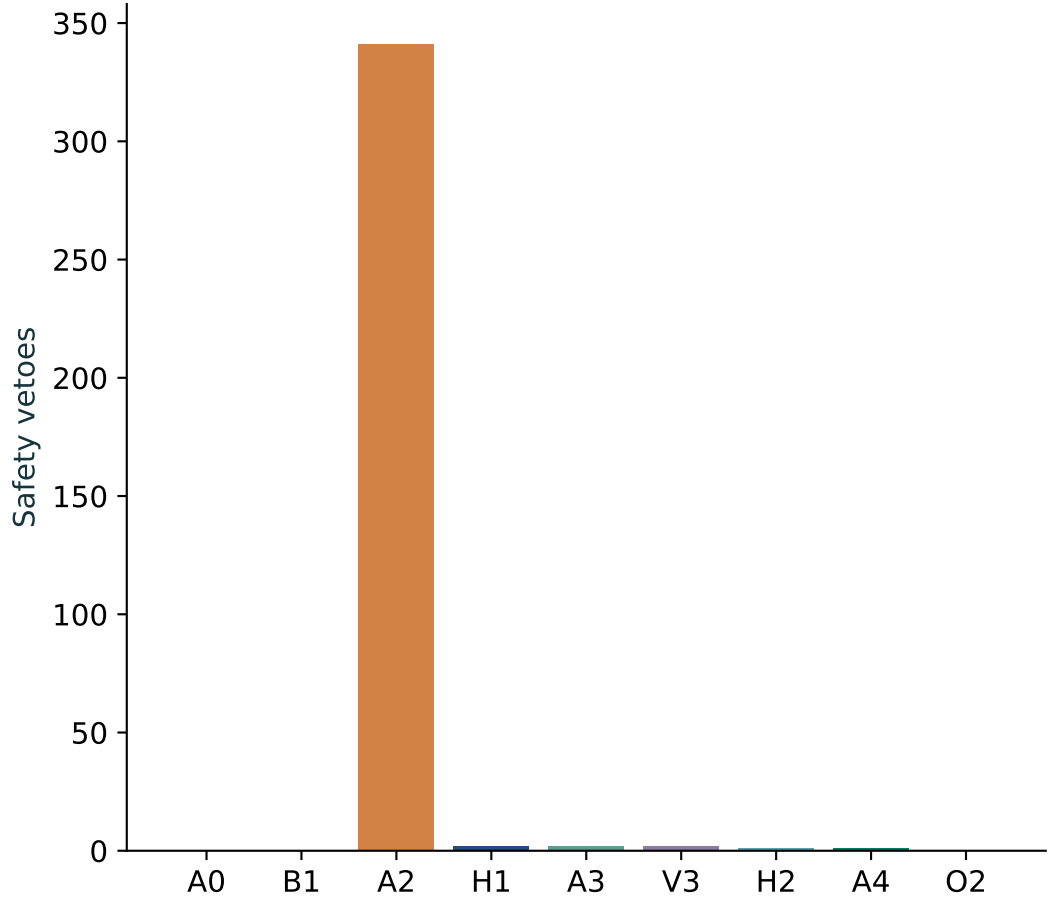




9,600 windows per controller, including overload, transitions and cold start. Individual windows within a seed are dependent; these curves are descriptive.

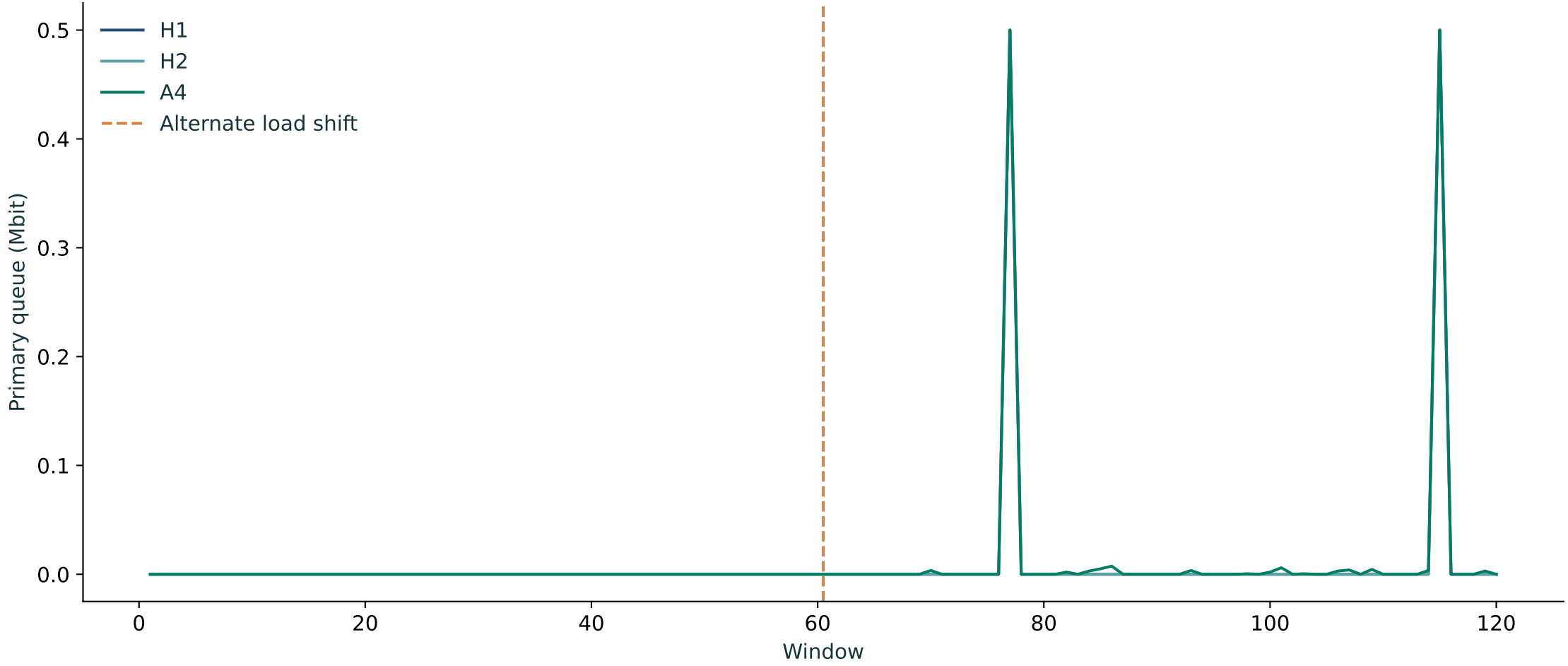
Installed actions and independent safety checks





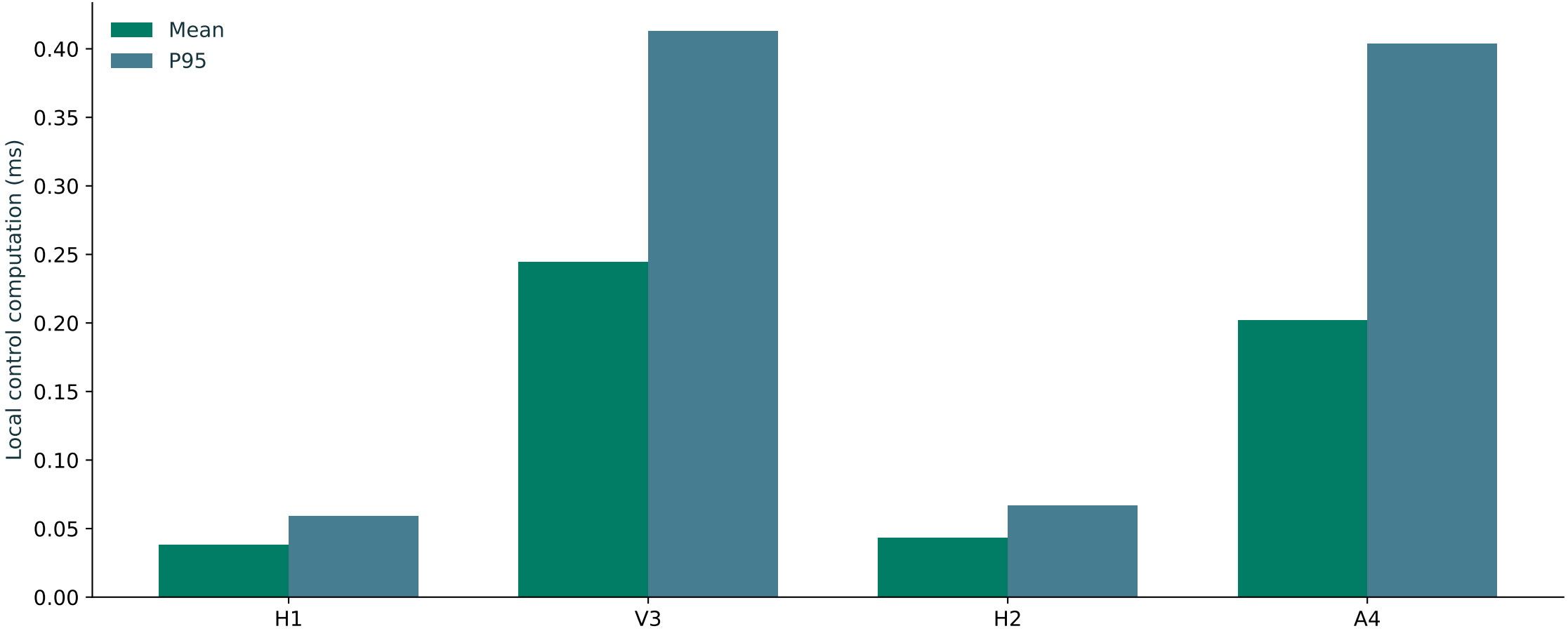
The unchanged safety limits enforce exact target/route, alternate utilization ≤90%, rate reduction ≤20% and minimum meter 512 kbps.

Queue response through a traffic distribution shift



Seed 1900 shown. Queues carry between windows; changing a rule does not erase queued traffic. Each intervention expires after five virtual seconds.

Actual local computation in the isolated .venv



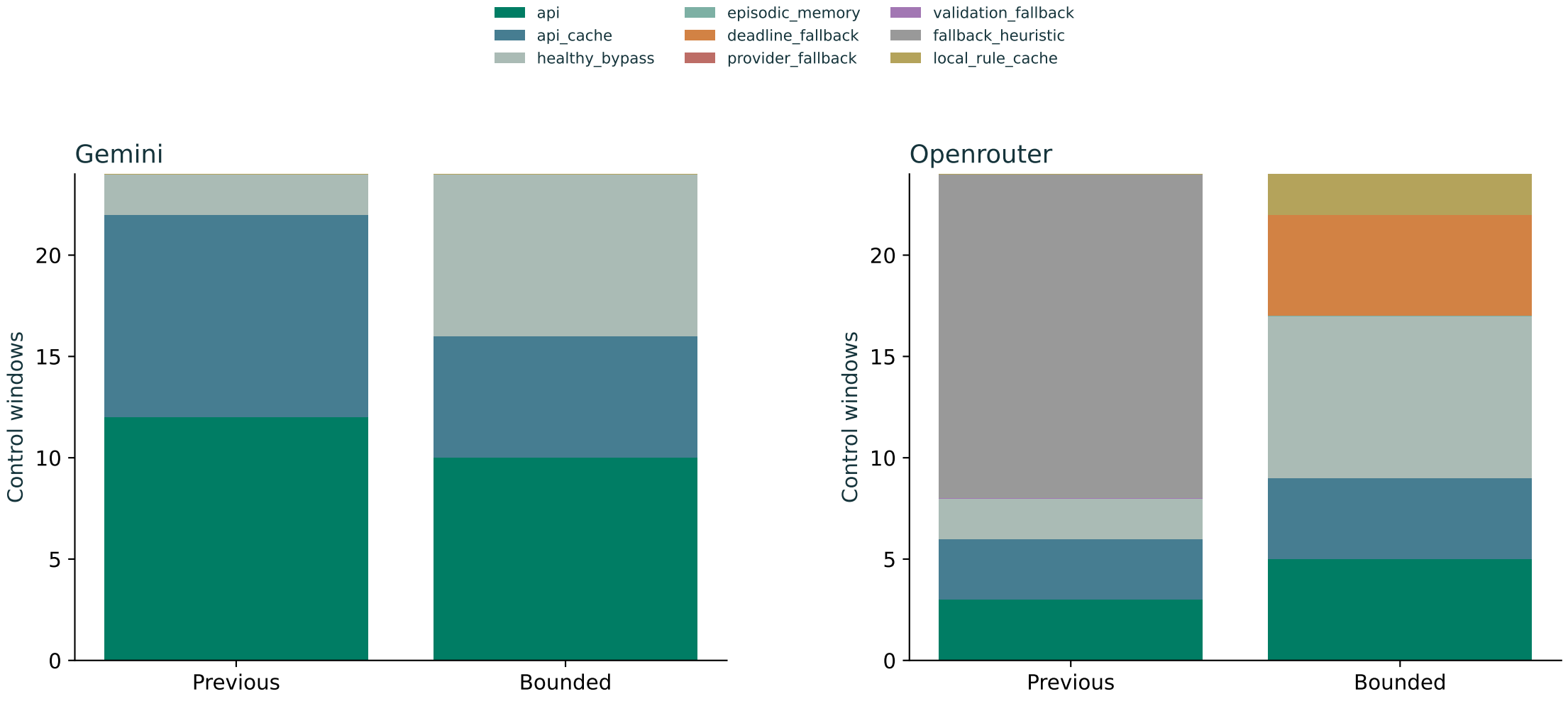
Per-window time includes selection, safety, fluid outcome, reward and memory update. Final batched memory flush excluded. Local timings depend on this machine and concurrent load.

Measured provider integration with five-second traffic arrivals



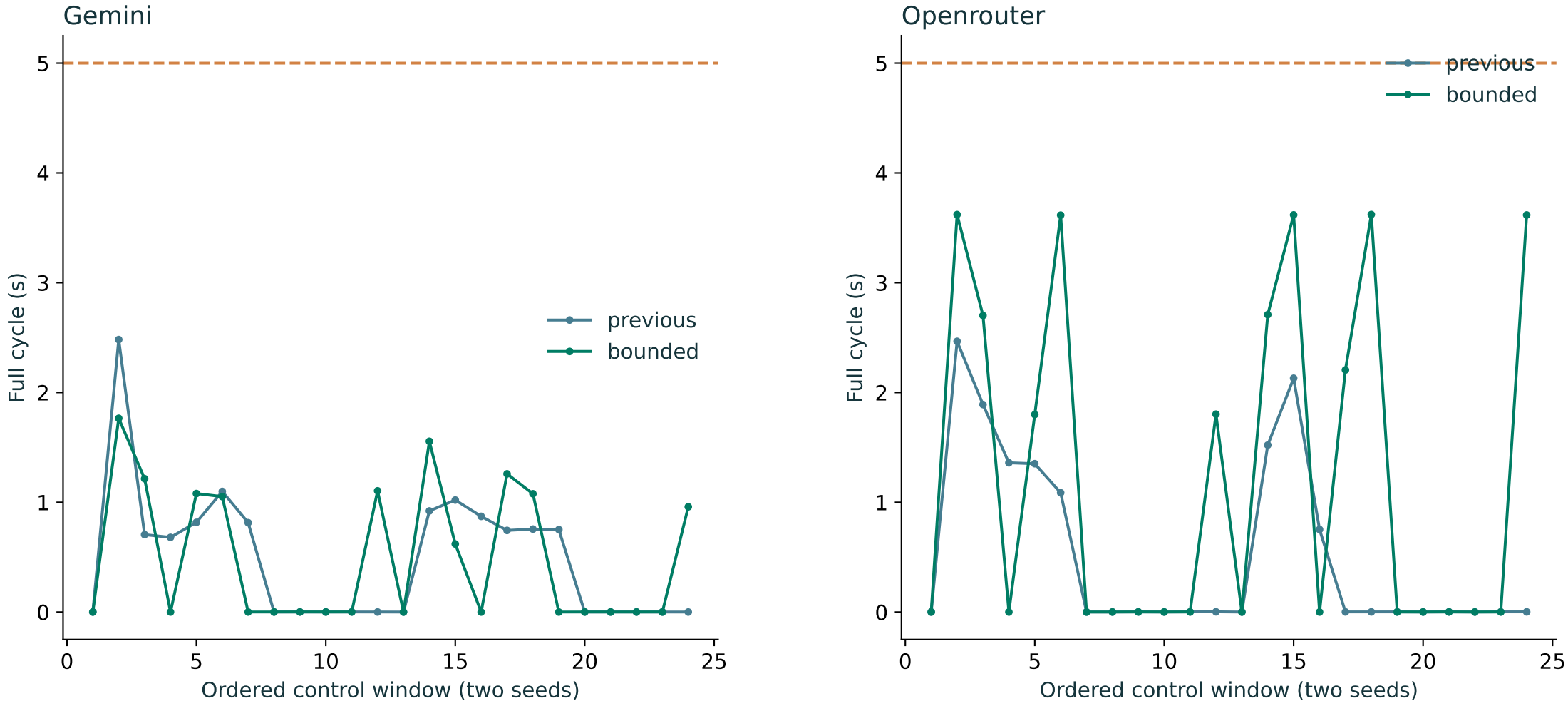
Two fresh seeds, 24 windows per mode/provider. Observation idle time is separately recorded; all control work, pacing, process startup, cancellation and cleanup is included. Timeouts remain in metrics.

Fresh suggestions, local reuse and fallbacks are separate



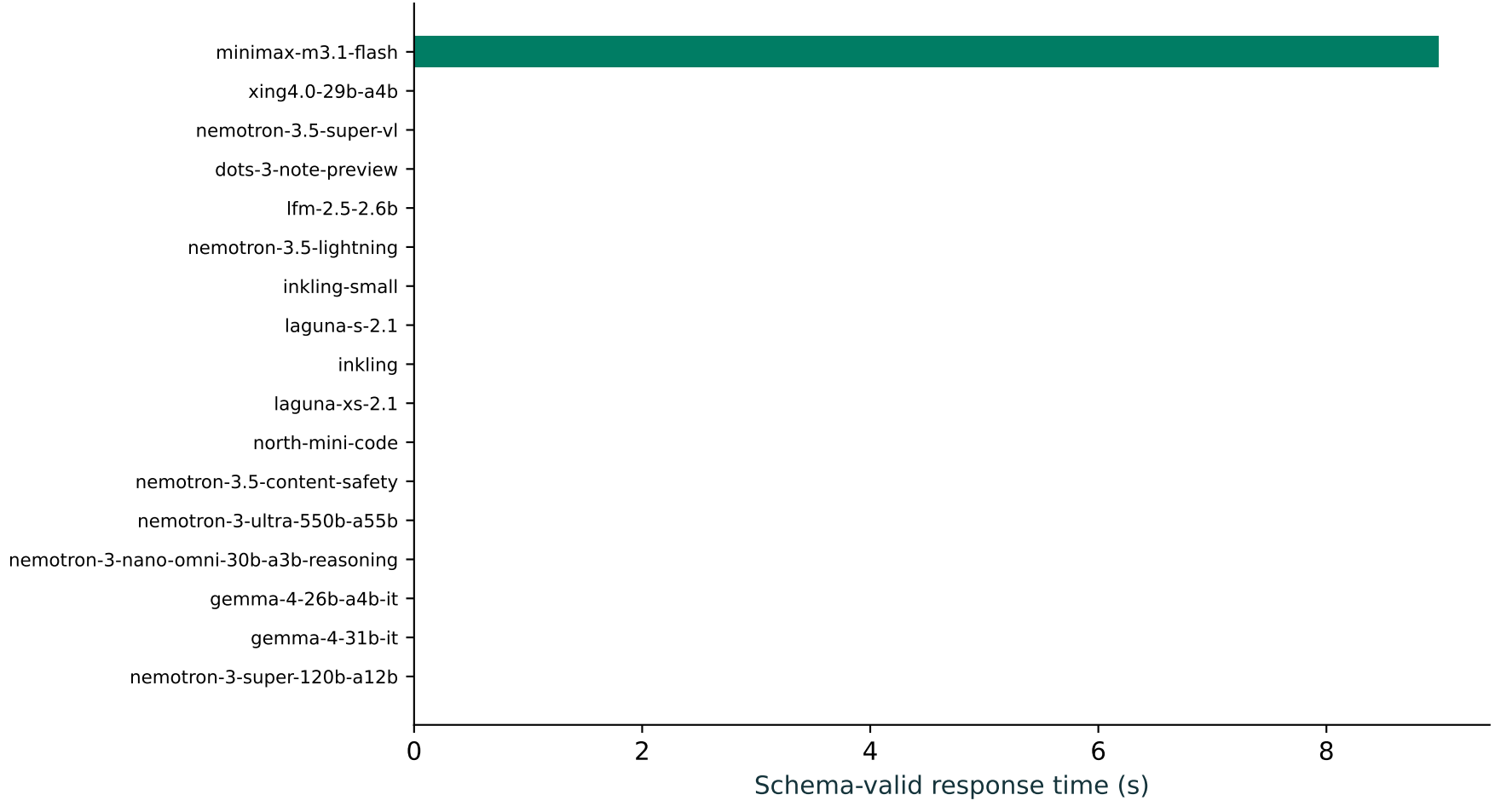
A provider timeout is a local client deadline, not a successful response. Admission deferrals start no request. No API work is silently moved outside the control interval.

Control tail latency with all windows retained



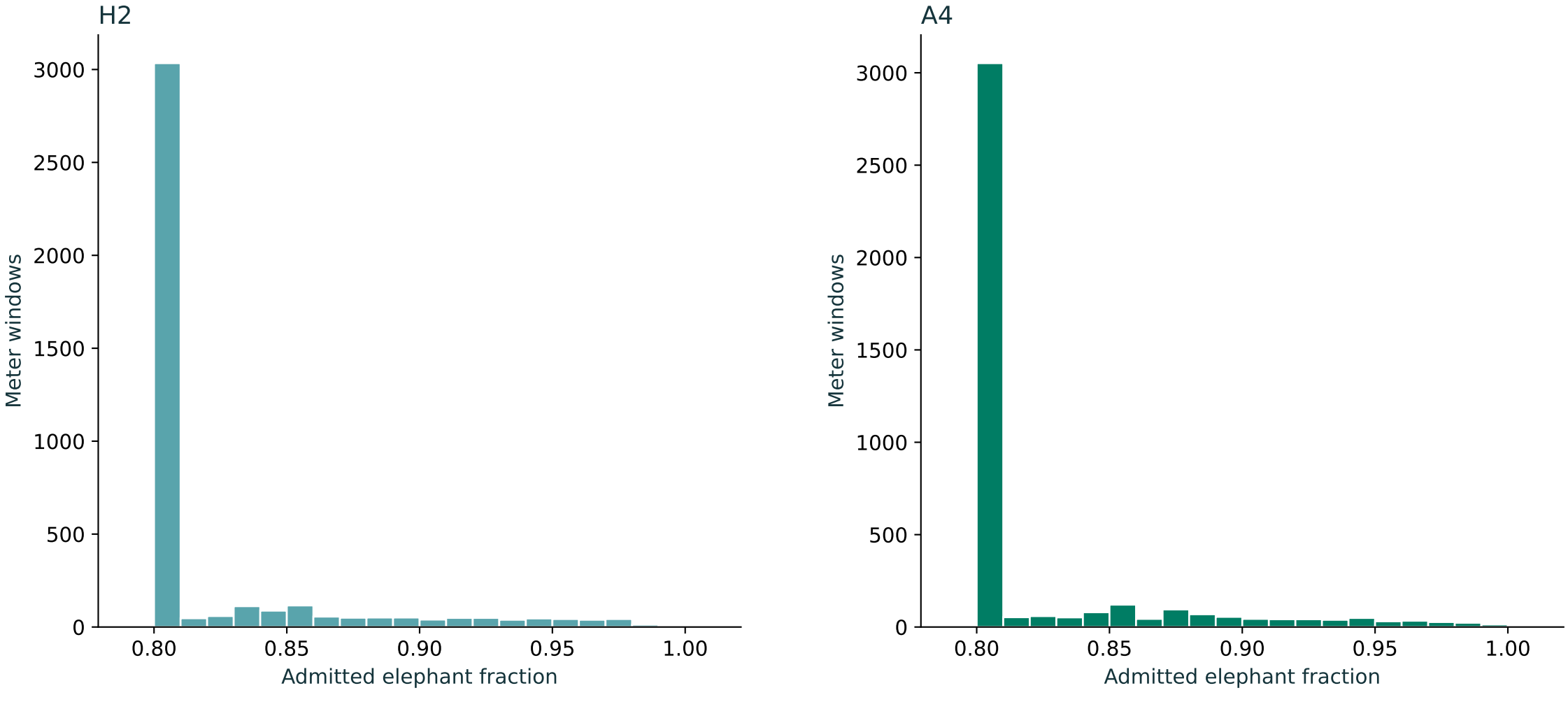
Modes use reversed order on the second seed to reduce order bias. Provider variability remains. A 3.8-second local wait budget leaves time for validation and measurement; it is not a hard real-time OS guarantee.

Availability of every discovered zero-price OpenRouter model



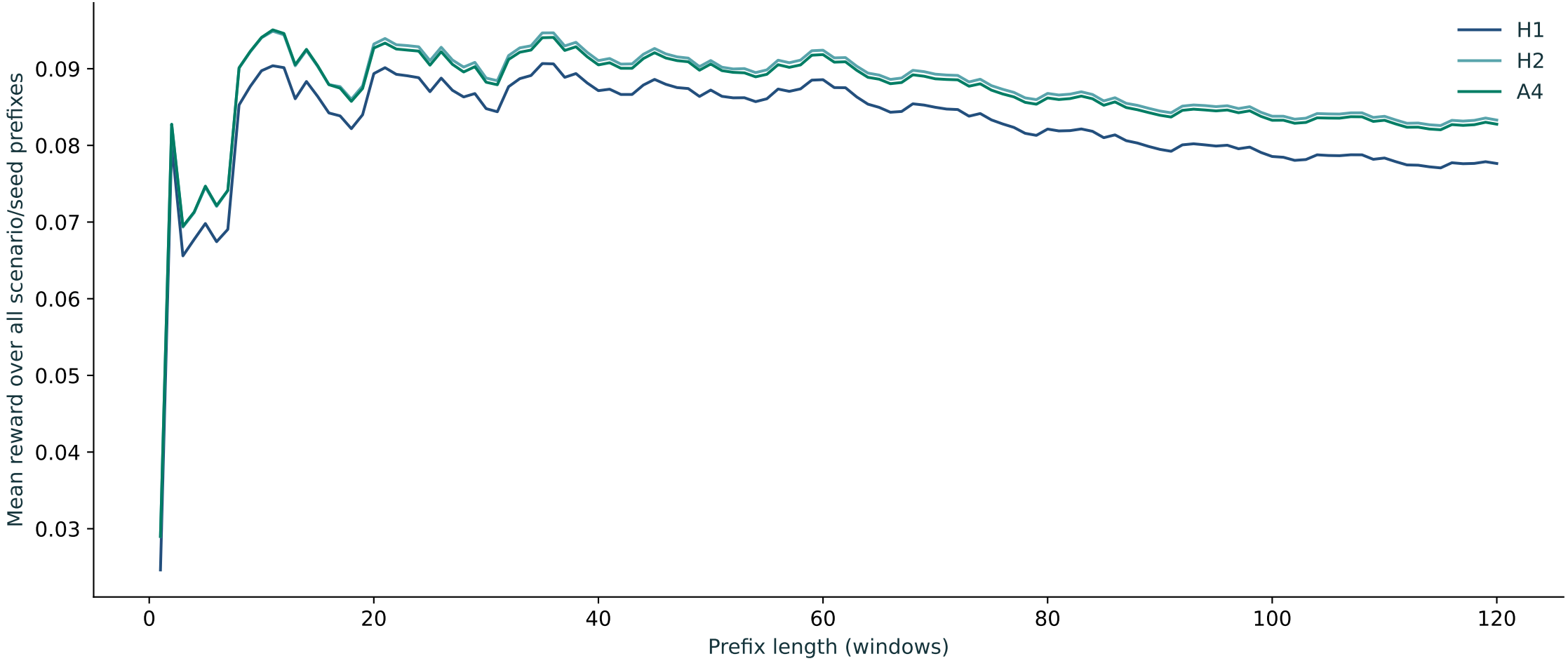
1/17 healthy-network schema checks valid. Zero bar marks unavailable/error, not zero latency. Availability check is separate from network-control quality or provider ranking.

Meter parameters retain more traffic than the fixed 80% rule



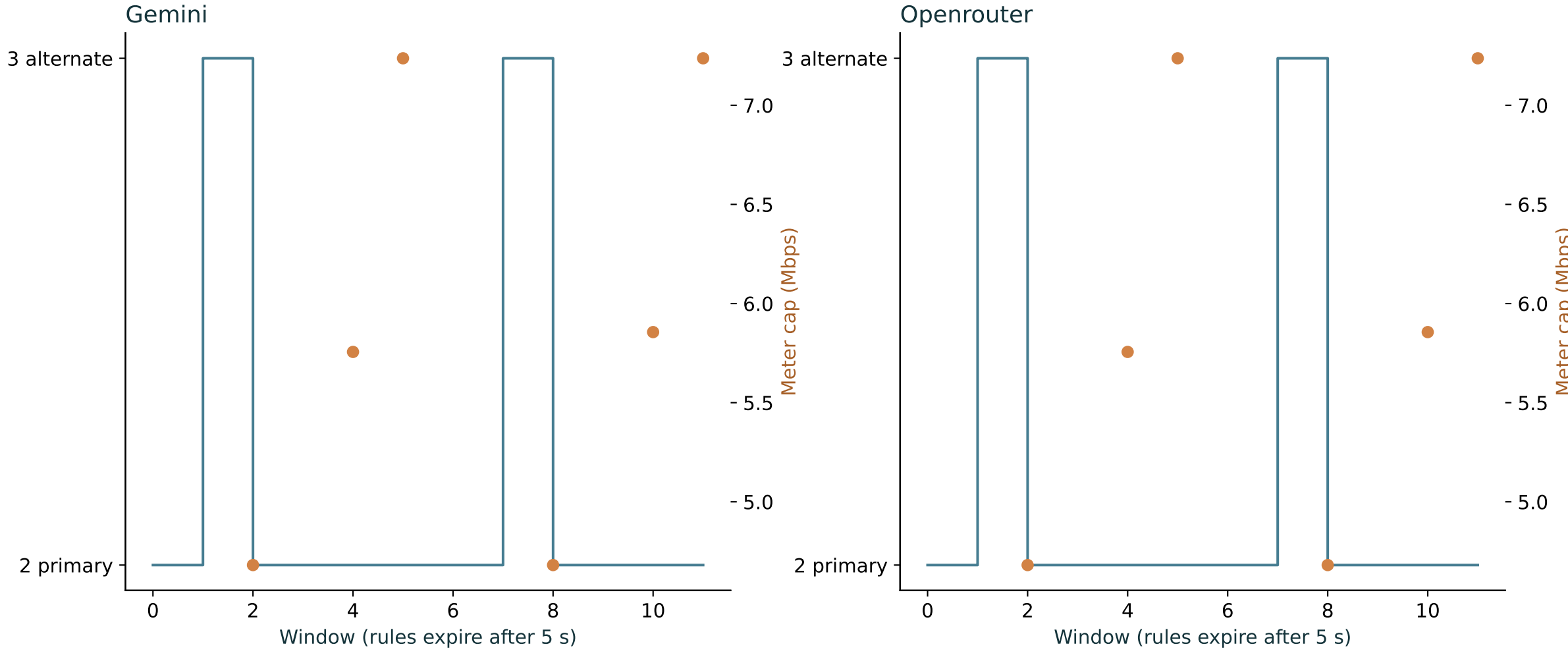
H2 uses a capacity-fitting meter with 0.1 Mbps headroom. A4 varies headroom through bounded prior-outcome exploration. The safety floor clips all candidates at at least 80% of elephant demand.

Memory learning includes its cold-start exploration cost



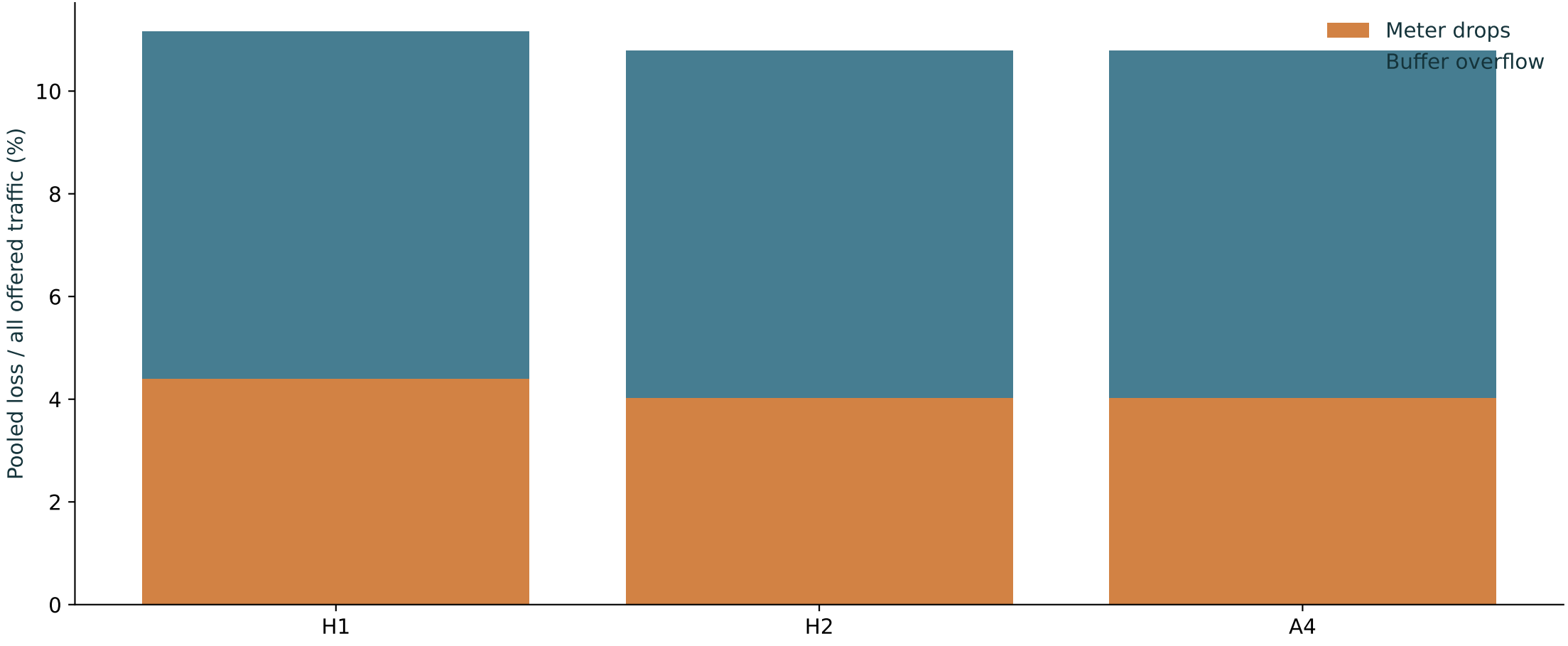
Primary evaluation, all eight scenarios and ten seeds. Higher reward is better. H2 is the matched no-memory ablation; an A4 advantage over H1 alone cannot prove an extra memory benefit.

Complete simulated rule changes: port, meter and expiry



Saved real-provider and fallback decisions drive a simulated OpenFlow table. Exact match, cookie, instructions, meter cap, prior-rule expiry and final default restoration are recorded. No physical switch is contacted.

Loss accounting keeps deliberate drops visible



Pooled volume loss differs from the arithmetic mean of window loss reported in primary tables. Conservation: prior queues + offered = delivered + all dropped + final queues.